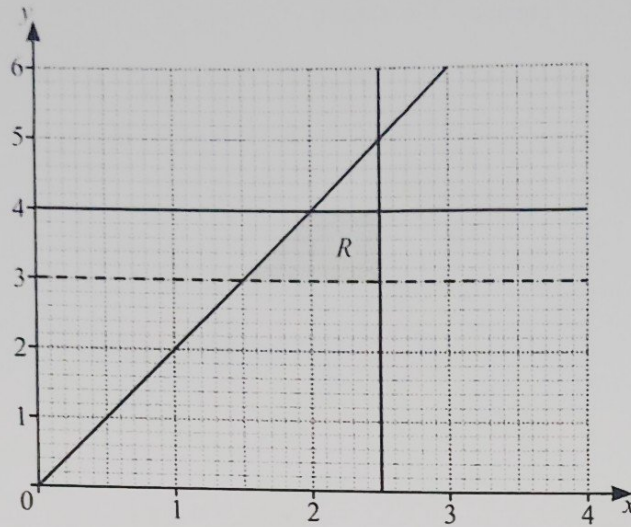


16



Write down all the inequalities that define the region R .

..... $y \geq x$
 $y \leq 4$
 $x \leq 2.5$
 $y \geq 3$
 *

[4]

17 $I = M(k^2 + c^2)$

(a) Find the value of I when $M = 7$, $k = 3$ and $c = 2$.

$I = 7(3^2 + 2^2)$ $I = 91$

$I = 7(13)$

$I = 91$ [2]

(b) Rearrange the formula to write k in terms of I , M and c .

$\frac{I}{M} = k^2 + c^2$

$\frac{I}{M} - c^2 = k^2$

$\sqrt{\frac{I}{M} - c^2} = k$

$k = \sqrt{\frac{I}{M} - c^2}$ [3]